Objectives and Scope

Section 1 — Objectives & Success Criteria

Primary Objective: Produce a data-driven comparative study of fixed-time signal vs. roundabout control at a single intersection, across a range of traffic volumes — not a verdict on which is universally better.

Supporting Objectives

#	Objective	Why
1	Realistic vehicle behavior (shared physics model)	Bad physics → meaningless comparison downstream
2	Identical test conditions for both strategies	Makes the comparison fair, not two unrelated sims
3	Fixed, precisely-defined metric set	Comparison is only as trustworthy as its measurement consistency
4	Visual + numeric output	Numbers alone are easy to distrust/misread
5	Repeatable, parameterized runs	One hardcoded scenario proves nothing generalizable

Success Criteria

#	Criterion	Pass condition
1	Both strategies run under identical, user-defined inputs	✅ / ❌
2	Full metric set computed consistently for both	✅ / ❌
3	Volume sweep produces genuinely different, interpretable results	Not flat/trivial across sweep
4	Non-technical viewer understands *what* changed and *why*	No code-reading required

Section 2 — Scope

In-Scope

Included	Why
Single, isolated intersection	No upstream/downstream effects muddying attribution
Motor vehicles only	Sufficient to show the core signal-vs-roundabout difference
2 strategies: fixed-time signal, gap-acceptance roundabout	The two strategies under study
Realistic car-following physics (IDM)	Differences must come from control logic, not scripted movement
Configurable volume, lane count, geometry	Primary levers that plausibly shift which strategy wins
Fixed, well-defined metric set (10 metrics, Section 7)	Precision = trustworthiness
Visual playback	Interpretable to non-technical viewers
Parameter sweep / batch mode across volumes	Produces a range, not one anecdotal data point
Repeatable runs, adjustable inputs	Tool must be explorable, not a one-off demo

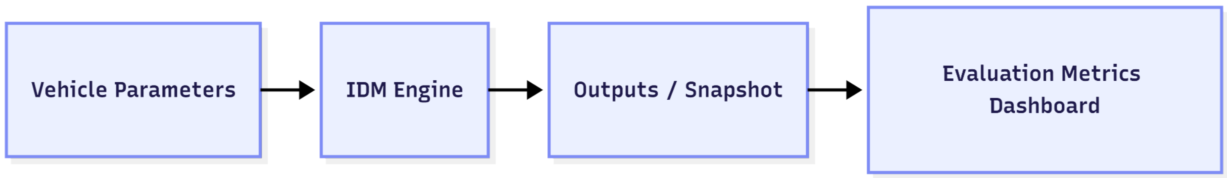
Out-of-Scope

Excluded	Why
Pedestrians, cyclists, multi-modal conflicts	Adds complexity without changing the core question
Emergency vehicle preemption	Real feature, but orthogonal to this comparison
Multi-intersection / corridor simulation	Different (larger) problem; breaks attribution
Full engineering-grade validation vs. real-world data	Comparative exploration tool, not a research-grade model

Declaring one strategy universally "better"	Deliverable is comparative data, not a verdict (see Master Formula caveats, Section 7.5)
Large extra-feature surface (hybrid models, NL input, etc.)	Prioritize few modules done well over many done shallowly

System Architecture and Modules

Section 3 — System Architecture



Component	Owns	Does NOT own	Repo location
Simulation Engine	Vehicle physics, control logic, metric computation	Rendering, display formatting	backend/ (Python + FastAPI)
Visualization & Analytics	Rendering, user input, statistical presentation	Vehicle simulation internals	frontend/ (React + TS + Vite)
Shared Contract Layer	Data schemas both sides must agree on	Business logic	shared/ (JSON Schema)

Key design decisions

Decision	Mechanism	Why
Snapshot contract	Fixed structure emitted every time step: vehicle positions, velocities, states, running metric counters	One thing to agree on before either side builds — frontend never recomputes metrics, only displays them
Single shared physics model (IDM)	All vehicles, both strategies, same car-following equation (Section 5)	Vehicle behavior is never a variable — only control logic differs between runs
Pluggable controller interface	SignalController / RoundaboutController both implement one interface (mayProceed, step) — sim loop never branches on which is active	Structurally impossible to favor one strategy in the loop itself
Fidelity: moderate, not engineering-grade	Real car-following + lane logic; no driver psychology variance, weather, multi-class vehicles	Enough realism for directionally trustworthy results without a multi-month build

Parameterization from day one	Volume, lane count, geometry are engine inputs, not hardcoded constants	Required for the sweep (Module 5) and user-adjustable controls (Module 8) — can't be retrofitted

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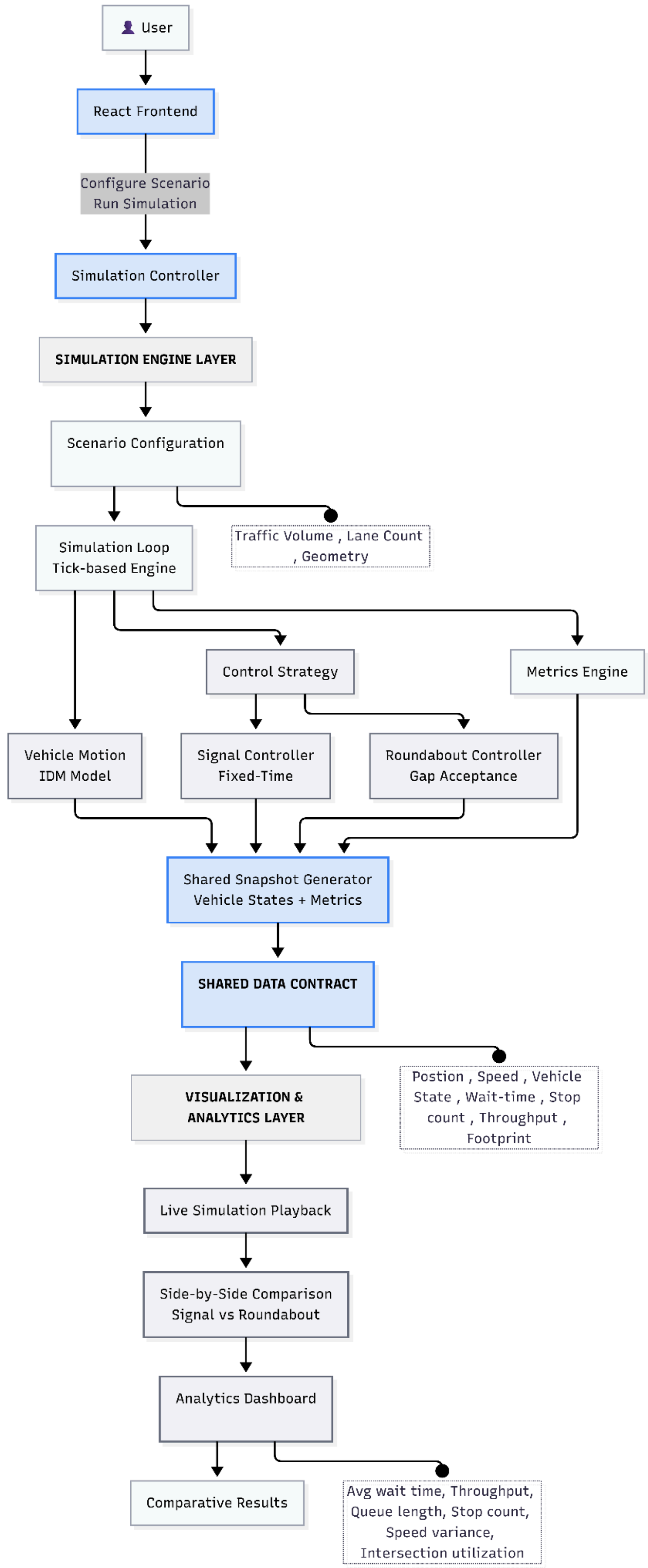
User
React Frontend
Configure Scenario
Run Simulation
Simulation Controller
SIMULATION ENGINE LAYER
Scenario Configuration
Traffic Volume , Lane Count , Geometry
Simulation Loop
Tick-based Engine
Control Strategy
Metrics Engine
Vehicle Motion
IDM Model
Signal Controller
Fixed-Time
Roundabout Controller
Gap Acceptance
Shared Snapshot Generator
Vehicle States + Metrics
SHARED DATA CONTRACT
Postion , Speed , Vehicle State , Wait-time , Stop count , Throughput , Footprint
VISUALIZATION &
ANALYTICS LAYER
Live Simulation Playback
Side-by-Side Comparison
Signal vs Roundabout
Analytics Dashboard
Comparative Results
Avg wait time, Throughput, Queue length, Stop count, Speed variance, Intersection utilization

Section 4 — Module Breakdown

#	Module	Function	Depends on	Why essential
1	**Vehicle Motion Engine**	Car-following physics (accel/brake/gap-keeping), strategy-independent	**—**	Foundation — if motion is inconsistent, nothing downstream is trustworthy
2	**Control Strategy Logic**	SignalController (fixed-time, phase-based) + RoundaboutController (gap-acceptance) behind shared interface	**Module 1**	This module **is** the variable under study
3	**Simulation Loop / Scenario Runner**	Steps sim forward, applies Modules 1–2, emits snapshots; runs a scenario end-to-end	**Modules 1, 2**	Guarantees "same scenario, different strategy" is mechanically true
4	**Metrics Computation**	Computes all 10 metrics (Section 7) identically regardless of active strategy	**Module 3**	Highest-risk module for silent bugs — inconsistent metric logic invalidates the whole study without being visually obvious
5	**Parameter Sweep / Batch Comparison**	Runs same scenario across a volume range, collects metrics at each level, both strategies	**Modules 3, 4**	Turns one anecdote into a study — this is the actual comparative dataset
6	**Visual Playback**	Renders vehicle movement + control state, ideally side-by-side	**Module 3 (snapshot)**	Grounds the metrics in observable cause/effect — builds trust, not just decoration
7	**Analytics & Comparison Output**	Tables/charts of Module 4+5 output across swept volumes	**Modules 4, 5**	The literal deliverable — "here's how they compared, here's how that changed with volume"
8	**Parameterized User Input**	UI/config to set volume, lanes, geometry pre-run	**Module 3**	Without it the tool is a demo, not an exploration instrument

Vehicle Motion Model (Intelligent Dri…

Section 5 — Vehicle Motion Model (Intelligent Driver Model)

Why IDM: widely validated car-following model (Treiber, Hennecke & Helbing, 2000). Produces ***smooth, non-discontinuous acceleration/braking*** with few, interpretable parameters — and it's ***control-strategy-agnostic by construction***, which is exactly what keeps vehicle ***behavior identical*** across SignalController and RoundaboutController runs (Section 3).

Governing equation:

$$\dot{v} = a\left[1 - \left(\frac{v}{v_0}\right)^{\delta} - \left(\frac{s^*(v,\Delta v)}{s}\right)^2\right]$$

$$s^*(v,\Delta v) = s_0 + vT + \frac{v\,\Delta v}{2\sqrt{ab}}$$

IDM basically uses this formula to calculate the acceleration/deceleration the vehicle should have at a particular instance. The first part $\left[1 - \left(\frac{v}{v_0}\right)^{\delta}\right.$ calculates **"free-road acceleration tendency"** and the second part of the equation $\left(\frac{s^*(v,\Delta v)}{s}\right)^2$ **"the desired dynamic gap".**

Parameters

Symbol	Name	Typical value	Meaning
v_0	Desired speed	13.9 m/s (~50 km/h)	Free-flow speed with no vehicle ahead
a	Max acceleration	1.0–1.5 m/s²	How eagerly the vehicle accelerates toward v_0
b	Comfortable braking decel.	1.5–2.0 m/s²	Normal (non-emergency) braking rate
δ	Acceleration exponent	4	Sharpness of accel taper as $v \to v_0$
s_0	Minimum jam gap	~2 m	Gap maintained at full standstill
T	Safe time headway	~1.5 s	Desired following gap, scales with speed
s	Actual gap	computed/step	Bumper-to-bumper distance to vehicle ahead
Δv	Approach rate	computed/step	$v - v_{lead}$; positive = closing in

How it applies to intersection control — "virtual lead vehicle" technique:

IDM only describes ***vehicle-to-vehicle*** following by itself. To make a vehicle stop at a signal or yield at a roundabout, a virtual stationary lead vehicle is placed at the stop/yield point whenever Controller.mayProceed(vehicle) returns false:

Controller says	IDM inputs used
mayProceed = false	s = distance to stop/yield point, $\Delta v = v - 0$
mayProceed = true	Real gap to actual vehicle ahead (or free-flow if none)

This is the mechanism that keeps ***Module 2 (control logic) fully decoupled from Module 1 (motion)*** — the controller only ever answers yes/no; IDM handles all ***acceleration/braking uniformly*** regardless of ***why*** a stop point exists.

Edge cases to handle explicitly

Case	Handling
s→0s or negative	Clamp to small positive ϵ\epsilon ϵ before computing s∗/ss^*/s s∗/s — avoids divide-by-zero spike
Vehicle stopped exactly at s0	Acceleration must evaluate ≈0, not oscillate (verify in POC 1)
Δv<0(lead pulling away)	Anticipatory term goes negative — must NOT clamp to zero, or "closing gap when lead speeds up" behavior is lost

Control Strategy Logic

Section 6 — Control Strategy Logic

Shared interface (both strategies implement this — nothing else touches vehicle state):

```
interface Controller {
  mayProceed(vehicle, simTime) -> bool   // pure query, no side effects
  step(simTime, dt) -> void              // mutates ONLY controller's own state
}
```

Contract guarantees (non-negotiable — these are what make the comparison fair):

Rule	Why
mayProceed is a pure function — never sets vehicle velocity/position directly	Only Module 1 (IDM) is allowed to turn a yes/no into actual motion
step mutates only the controller's own state (e.g. phase timer)	Prevents either controller from reaching into vehicle/other state
Both queryable per-approach / per-vehicle	Signal phase applies per approach; roundabout yield is evaluated per vehicle against specific conflicting traffic
step always runs before any mayProceed calls that step	A phase transition mid-step must apply before decisions are made, for both controllers equally

---

6.1 — SignalController (Fixed-Time)

Model	Deterministic FSM cycling fixed phases (NS_Green → NS_Yellow → AllRed → EW_Green → EW_Yellow → AllRed)
step()	Increments phase timer; advances to next phase when duration elapsed
mayProceed()	true only if vehicle's approach has Green in the current phase
Defining property	Phase duration is **independent of queue length or arrival rate** — no adaptive/actuated logic. This is the deliberate contrast point against the roundabout; adding actuation would blur the comparison and fall outside scope (Section 2).
Yellow / All-Red	mayProceed = false for all approaches, except a configurable "dilemma zone" carve-out: vehicles already within X meters of the stop line at Yellow onset may proceed

6.2 — RoundaboutController (Gap-Acceptance)

Based on standard roundabout capacity theory (HCM / Tanner-style gap acceptance).

Concept	Symbol	Meaning	Typical value
Critical gap	tc	Min. circulating-traffic gap a driver accepts	4.0–4.5 s
Follow-up time	tf	Min. spacing between successive entries on one gap	2.5–3.0 s

Available gap:

$$t_{gap} = \frac{d_{circ}}{v_{circ}}$$

(d_{circ} = distance of nearest circulating vehicle to conflict point, v_{circ} = its speed)

Decision rule:

$$\text{mayProceed} = \begin{cases} \text{true} & \text{if } t_{gap} \geq t_c \text{ and } (\text{simTime} - t_{lastEntry}) \geq t_f \\ \text{false} & \text{otherwise} \end{cases}$$

Case	Handling
No circulating vehicle within horizon	Tgap to ∞ → mayProceed = true immediately. This is the mechanism behind near-zero delay at light traffic — the project's core thesis.
Multi-approach roundabout	Circulating-flow state tracked **per conflict point**, not globally — each approach yields independently
step()	Does **no** proactive time-based advancement (unlike the signal's phase timer) — only updates lastEntryTime reactively, inside mayProceed, at the moment entry is granted

6.3 — Why the split is fair, not just convenient

Both controllers answer the identical question through the identical interface, consumed by the identical motion model (Section 5, virtual-lead-vehicle mechanism). The *only* asymmetry is what information each uses to decide: a fixed clock (signal) vs. live circulating-traffic state (roundabout) — which is exactly the real-world distinction under study, not an artifact of implementation.

(Full multi-lane extension notes, dilemma-zone parameter details: Appendix B.2)

Data & Simulation History Storage

Section 7 — Data & Simulation History Storage

Aspect	Implementation
Storage	• Local **SQLite database** on the user's machine • No separate database server required
Why	● **Simple & lightweight** — no separate database server or complex setup. ● **Local & portable** — entire database can be stored as a single file with the application. ● **Sufficient for project scale** — easily handles simulation runs, snapshots, and metrics. ● **Easy history retrieval** — supports storing and querying past runs without rerunning simulations.
Run Information	• Unique Run ID • Timestamp • Signal / Roundabout strategy • Run status
Scenario Data	• Traffic volume • Lane count • Geometry • Vehicle parameters • Random seed
Snapshots	• Simulation timestamp • Vehicle position • Speed • Vehicle state
Metrics	• Average wait time • Average travel spee • Stop count • Throughput • Queue length
Data Flow	**Run → Snapshots → Metrics → SQLite → Dashboard**
History Access	• Select previous Run ID • Load stored snapshots/metrics

	• View or compare results without rerunning
Purpose	• Persistent history • Faster result access • Reproducible runs • Historical comparison

Evaluation Metrics

Section 7 — Evaluation Metrics

The simulation will compare the fixed-time signal and roundabout under the same traffic demand, road geometry, vehicle model, and simulation duration.

The metrics below are calculated using the same definitions for both strategies. Standard traffic-engineering concepts are taken from FHWA/HCM guidance; where no single standard formula exists, the formula is explicitly defined for this project.

The same metrics and calculation rules will be applied to both **fixed-time signal** and **roundabout** simulations under identical conditions.

Metric Purpose

Category	Main Question
Operational Efficiency	How efficiently does the intersection process traffic?
Traffic Flow Quality	How smoothly and consistently do vehicles move?
System Performance	How effectively is available capacity being used?
Fairness & Stability	Are delays balanced and congestion stable?
Physical Constraints	How much physical space does the design require?

7.1 Operational Efficiency

Metric	Definition & Formula	Threshold / Rule	Reference
Average Wait Time	Average time a completed vehicle spends stopped/crawling in the intersection zone. **AWT = Σ vehicle wait time / completed vehicles**	Vehicle considered waiting when **$v < 1$ m/s**. Final threshold will be fixed before testing.	FHWA Traffic Signal Timing Manual
Throughput	Number of vehicles that completely clear the intersection per hour. **Throughput = Vehicles cleared / Measurement time × 3600**	Count only vehicles that cross the defined exit boundary.	FHWA / HCM
Queue Length	Number of vehicles waiting at each simulation step. Report **Average, Maximum and P95 queue length**.	Vehicle enters queue when **$v < 1$ m/s** within the influence zone.	FHWA / HCM

7.2 Traffic Flow Quality

Metric	Definition & Formula	Threshold / Rule	Reference
Stop Count	Number of separate stopping events per vehicle. **Average Stops = Total stops / completed vehicles**	A stop occurs when speed changes from **≥ 1 m/s to < 1 m/s**. Continuous stopping counts once.	FHWA Traffic Signal Timing Manual
Speed Variance	Measures variation in vehicle speed. **$Var(v) = \Sigma(v_i - \bar{v})^2 / (n-1)$**	Calculate only inside the influence zone.	Project-defined
Travel Time Reliability	Measures how consistent vehicle travel time is. Report **Mean, SD and P95 travel time**.	**Travel Time = Exit time − Entry time**.	Transportation reliability methodology; project implementation
Average Travel Speed Through Intersection	Average speed of completed vehicles while traversing the defined intersection zone. **ATS = Σ (vehicle travel distance / vehicle travel time) / completed vehicles**	Same entry/exit boundaries for both strategies. Include only vehicles that completely clear the intersection.	FHWA Traffic Signal Timing Manual

7.3 System Performance

Metric	Definition & Formula	Threshold / Rule	Reference
Idle Opportunity Loss	Time when vehicles are waiting even though at least one vehicle could safely proceed.	Count only when **waiting vehicle + safe opportunity + no vehicle served** are all true.	Project-defined; based on control behaviour
Critical Saturation Volume	Traffic demand at which performance begins to degrade continuously.	Threshold determined from the volume sweep using predefined **delay/queue/throughput degradation criteria**.	Traffic-capacity concept; exact rule project-defined

Intersection Utilization	Percentage of active-demand time during which vehicles are being served. **Utilization = Service time / Active-demand time × 100**	Periods with no approaching/waiting vehicles are excluded.	Project-defined

7.4 Fairness & Stability

Metric	Definition & Formula	Threshold / Rule	Reference
Directional Fairness Index	Measures how evenly delay is distributed between approaches. **DFI = SD(approach average wait times)**	**Lower value = more balanced delay.**	Project-defined
Queue Stability Index	Measures queue fluctuation over time. **QSI = SD(queue length) / Mean(queue length)**	**Lower value = more stable queue.**	Project-defined
Congestion Recovery Time	Time required for the queue to return to normal after heavy congestion. **Recovery Time = Normal-condition time − Congestion-end time**	Normal-condition threshold will be fixed before testing.	Project-defined

7.5 Physical Constraints

Metric	Definition & Formula	Threshold / Rule	Reference
Space / Footprint Consumed	Physical area required by the configured intersection geometry. Examples: **$A = L \times W$**, **$A = \pi r^2$**.	Static metric; independent of traffic volume.	Project geometry model

Common Measurement Rules

Parameter	Definition
Influence Zone	Fixed road area around the intersection where performance is measured.

Crawl-Speed Threshold	Initial definition: **1 m/s**. Final value will be frozen before the main experiment.
Completed Vehicle	Vehicle that enters and fully exits the measurement zone.
Warm-up Period	Initial simulation period excluded to remove startup effects.
Measurement Period	Fixed period after warm-up used for final metric calculations.
Random Seed	Same seed and traffic demand used for corresponding signal and roundabout runs.

Execution Plan

Section 8 — Execution Plan for 2 months

Sprint	Duration	Sprint Goal / User Story	Viraj – Simulation / Data	Khushi – Frontend / Analytics	Working Increment / Exit Criterion
S1	Days 1–3	Establish runnable project baseline	Project setup + basic IDM single-vehicle model	Frontend shell + basic UI	**V0.1:** Basic vehicle simulation runs and is visible in UI.
S2	Days 4–7	Simulate multiple vehicles	Extend IDM and validate vehicle following	Render multiple vehicles	**V0.2:** Multiple vehicles move realistically.
S3	Days 8–11	Add signal control	Implement SignalController + common loop	Add signal state to UI	**V0.3:** Signal-controlled scenario runs and displays.
S4	Days 12–15	Add roundabout control	Implement RoundaboutController using same interface	Add roundabout playback	**V0.4:** Both strategies run through the same framework.
S5	Days 16–19	Connect real simulation data	Generate simulation snapshots	Display live vehicle states	**V0.5:** Real engine → frontend flow works.
S6	Days 20–23	Add evaluation metrics	Calculate wait time, speed, stops and throughput	Display metrics with playback	**V0.6:** Both strategies produce comparable metrics.
S7	Days 24–27	Enable direct comparison	Run identical scenarios with repeatable seeds	Side-by-side strategy comparison	**V0.7:** Signal vs. roundabout can be visually compared.

Sprint	Duration	Sprint Goal / User Story	Viraj – Simulation / Data	Khushi – Frontend / Analytics	Working Increment / Exit Criterion
S1	Days 1–3	Establish runnable project baseline	Project setup + basic IDM single-vehicle model	Frontend shell + basic UI	**V0.1:** Basic vehicle simulation runs and is visible in UI.
S8	Days 28–31	Add data storage & traffic-volume analysis	Store runs, snapshots and metrics in **local SQLite**; run volume sweeps	Build history view + comparison charts	**V0.8:** Previous runs can be retrieved without rerunning; comparison data is generated across traffic volumes.
S9	Days 32–35	Make simulation interactive	Add lane/geometry inputs, restart and validation	Add parameter controls and fresh-run flow	**V0.9:** Changed inputs produce fresh, correctly stored results.
S10	Days 36–40	Validate and prepare final study	Stability, edge cases, reproducibility and final data export	Dashboard polish, final comparisons and screenshots	**V1.0:** Complete manager-ready tool with validated comparative results.

Proof of Concepts

Section 9 — Proof of Concepts

Ordered so each POC only makes sense once the prior one holds — no validating metric consistency before the motion model itself is proven stable.

#	POC	Risk it addresses	Pass condition	Blocks if it fails
1	Vehicle Motion Stability	Glitchy/unrealistic car-following invalidates every downstream result	Plausible spacing, smooth accel/decel, no collisions/freezes across vehicle counts & speeds	Everything downstream
2	Shared Controller Interface	If both controllers can't share one interface without special-casing, the "fair comparison" architecture breaks structurally	Both controllers run the identical scenario through the identical loop code, zero errors, valid snapshots both times	Fair-comparison architecture (Section 3)
3	Metric Computation Consistency	A silent definitional inconsistency between strategies invalidates the study without being visually obvious	Hand-traceable small scenario (~5 vehicles) — manually computed metrics match engine output, both strategies	Credibility of every reported number
4	Meaningful Variation Across Volume	Flat results across volume = static demo, not a comparative study	Metrics visibly shift across a small volume sweep (low→near-saturation), for both strategies	Batch sweep / analytics build (before investing full effort there)
5	Synchronized Visual Playback	Drifted/cluttered playback destroys the visual layer's communicative value	Viewer can clearly distinguish waiting/yielding/moving state; both sides stay in sync throughout	Visual comparison value
6	Parameter Propagation Correctness	Stale state / silent no-ops make the tool untrustworthy as an exploration instrument	Changing one input (e.g. volume) → output shifts in expected direction, no leftover state from prior run	User-facing exploration trust

Risks & Assumptions, Future Scope

Section 10 — Risks & Assumptions

Risks:

Risk	Why it matters	Mitigation
Metric definitions drift between controllers over time	Silently invalidates the comparative study, not visibly obvious	POC 3 validates upfront; metric logic lives in one shared module (not duplicated per controller)
Moderate-fidelity physics looks reasonable but is subtly wrong at an untested edge case (e.g. extreme volume → unrealistic gridlock)	Misleading conclusion at extremes	Scope volume sweep to realistic range (light→near-saturation); flag anomalous results during review, don't auto-trust all output
Shared snapshot contract not finalized before both people start building	High rework cost if contract changes after both sides have built against it	Contract locked in Phase 0 before scenario-specific logic starts; daily check-ins early on watch for mismatches
Timeline slips (8-week plan is an estimate)	Stretch features could get squeezed, or worse, core modules get rushed	Stretch features explicitly cuttable (Section 11) so core is never what's sacrificed
Visual playback becomes a "nice demo" not backed by trustworthy metrics — or accurate metrics no one can interpret without it	Either failure undermines the tool's core communicative purpose	POC 5 explicitly tests that visual + numeric layers reinforce, not substitute for, each other

Assumptions:

#	Assumption
1	Volume, lane count, geometry are the primary levers; weather, driver variance, time-of-day are constant/out of scope
2	Moderate fidelity is sufficient for directionally trustworthy insight, even without validation against real-world datasets
3	Both members have or can quickly pick up their layer's technical background; no separate onboarding phase budgeted
4	The 8-week plan assumes roughly consistent availability from both people; not padded for extended unavailability

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Section 11 — Future Scope (Deliberately Deferred)

Deferred feature	Why compelling	Why deferred
Run history / persistence	Revisit past configs/results instead of losing on refresh	Usability convenience layer; zero impact on whether the core comparison is fair/clear/deep
Hybrid control model (3rd comparison column)	Extra data point	Third strategy = its own design + validation burden; core two-way comparison must be solid first
Natural-language parameter input	Lower-friction config	Stretch UX layer over Module 8 — sliders already satisfy the "explorable, parameterized" objective
~~User-weighted recommendation scoring~~	Personalized takeaway from raw metrics	**Reinstated** — see Master Efficiency Formula (Section 7A). Originally deferred over verdict-framing risk; now added at manager's request, with normalization/weighting caveats explicitly documented there

Additional metrics (emissions, fuel, conflict-point safety counts)	Real, meaningful comparison dimensions	Each needs its own modeling + validation effort disproportionate to marginal insight now
Edge-case traffic conditions (uneven directional volume, oversaturation stress-test)	Would stress-test both strategies further	Core light-to-heavy sweep sufficient to establish the primary finding first

References

References (single consolidated section — group by topic, place near end of doc / before Appendices)

Vehicle Motion Model

	Reference	Used for
1	Treiber, M., Hennecke, A., & Helbing, D. (2000). *Congested traffic states in empirical observations and microscopic simulations.* Physical Review E, 62(2), 1805–1824.	Intelligent Driver Model (IDM) — Section 5 governing equation

Roundabout Control Logic

	Reference	Used for
2	Transportation Research Board. *Highway Capacity Manual (HCM)* — Roundabout Methodology.	Gap-acceptance model basis, critical gap / follow-up time concepts — Section 6.2
3	Tanner, J.C. (1962). *A theoretical analysis of delays at an uncontrolled intersection.* Biometrika, 49(1/2), 163–170.	Underlying gap-acceptance theory (Tanner/HCM-style formulation) — Section 6.2

Evaluation Metrics — Operational Efficiency & Traffic Flow Quality

	Reference	Used for
4	Federal Highway Administration (FHWA). *Traffic Signal Timing Manual.* U.S. Department of Transportation.	Average Wait Time, Stop Count definitions — Section 7.1, 7.2
5	FHWA / Highway Capacity Manual (HCM)	Throughput, Queue Length definitions — Section 7.1
6	Transportation reliability methodology (general practice; project-specific implementation)	Travel Time Reliability — Section 7.2

Project-Defined (no external standard — explicitly flagged, not sourced)

	Item	Note
7	Speed Variance, Idle Opportunity Loss, Critical Saturation Volume, Intersection Utilization, Directional Fairness Index, Queue Stability Index, Congestion Recovery Time, Footprint	No single established formula exists for these in FHWA/HCM literature — definitions were authored for this project. Thresholds to be frozen before the final experiment (Section 7 note) so definitions aren't adjusted after results are seen.

Terminology

	Reference
8	FHWA = Federal Highway Administration, U.S. Department of Transportation — issuing body for refs #4 and #5